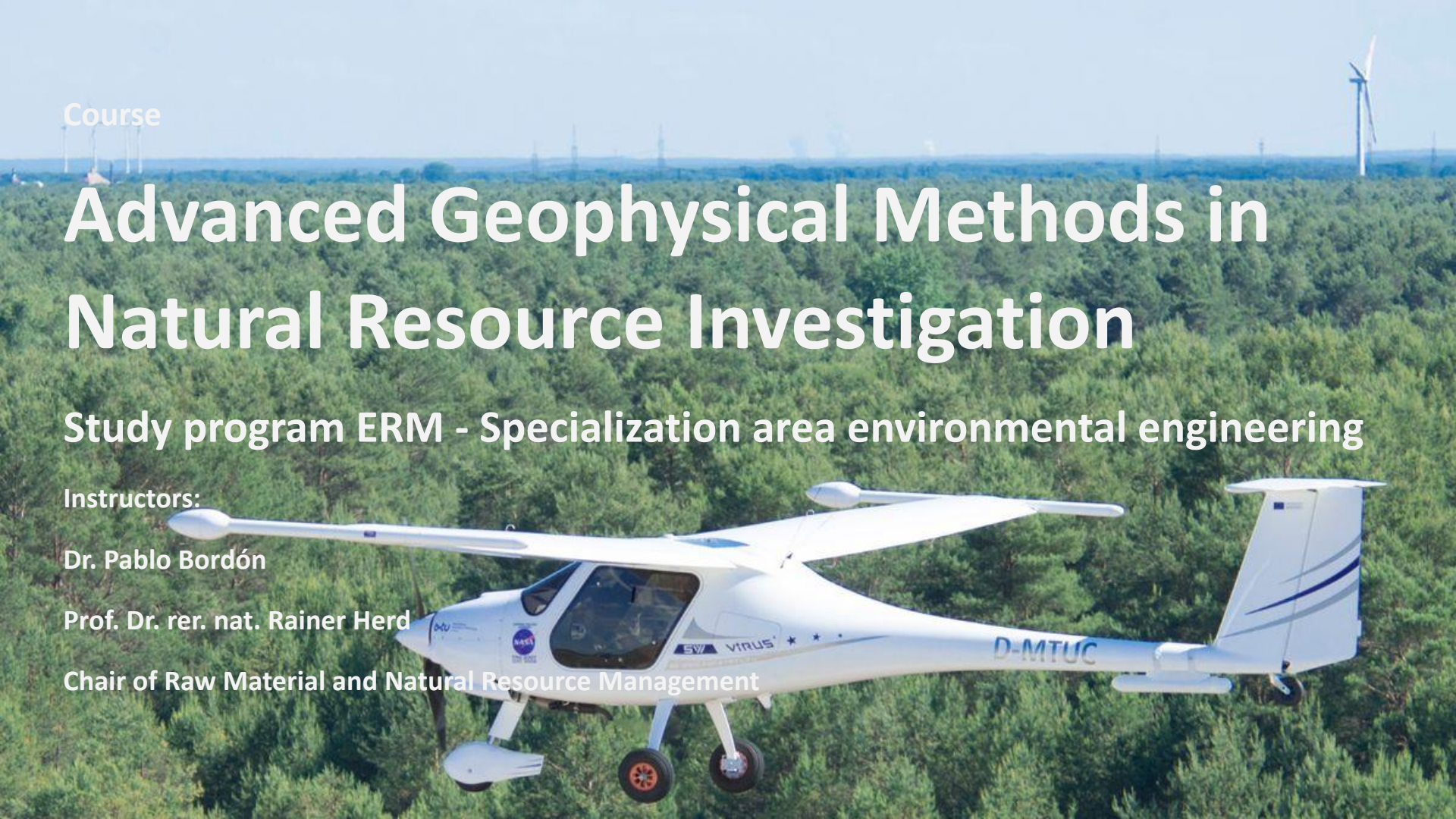
Course

Advanced Geophysical Methods in Natural Resource Investigation

Study program ERM - Specialization area environmental engineering



Instructors:

Dr. Pablo Bordón

Prof. Dr. rer. nat. Rainer Herd

Chair of Raw Material and Natural Resource Management

Course Overview

Duration: 1 semester

Credits: 6

Mandatory prerequisites: Successful completion of the module (*42421*) Natural Resource Investigation.

Limited number of participants: 15

Forms of teaching: Lectures + Field training + Self-study 120 (hours)

Assessment mode for module examination:

- Written exam covering the lecture content (90 minutes)
- Oral presentation of field and lab work (15 minutes)
- Written technical report for the practical training (10-15 pages)

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Course Overview

Objective:

- To apply geophysical prospecting methods for the exploration of natural resources and other related applications.

Course Overview

Objective:

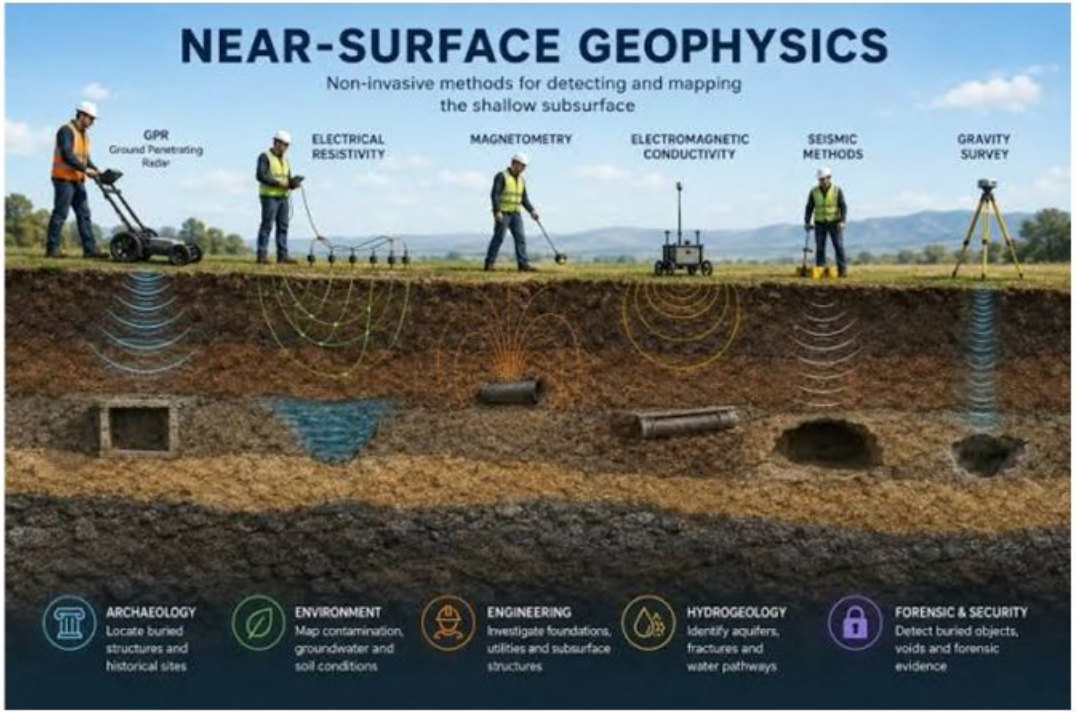
- To apply **geophysical prospecting methods** for the **exploration of natural resources** and other **related applications**.

Techniques that use physical principles (electrical, electromagnetic, magnetic, radiometric) to investigate the subsurface without excavation.

The search for economically valuable materials (e.g. groundwater, minerals) using technical and scientific approaches.

- Civil engineering (e.g. tunnels, foundations)
- Environmental assessments
- UXO detection
- Archaeological investigations

Near-surface geophysics → One toolbox, many problems



Fields of application ·

Groundwater & salinisation

Raw materials & critical minerals

Environment & contamination

Engineering & geohazards

Archaeology & cultural heritage

Course Overview

Objective:

- To apply **geophysical prospecting methods** for the **exploration of natural resources** and other **related applications**.

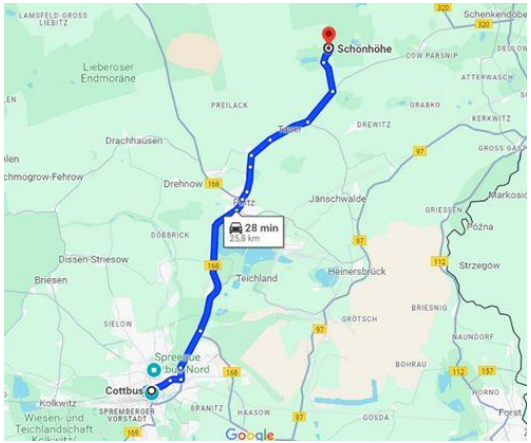
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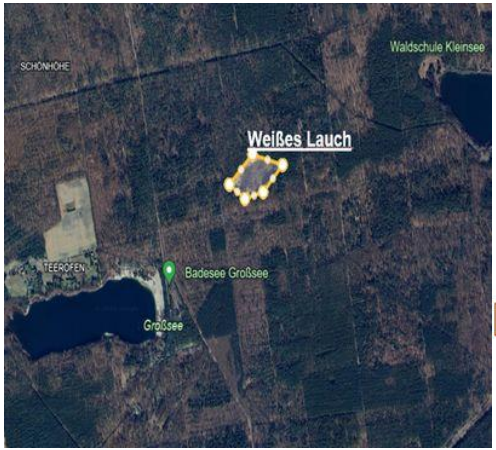
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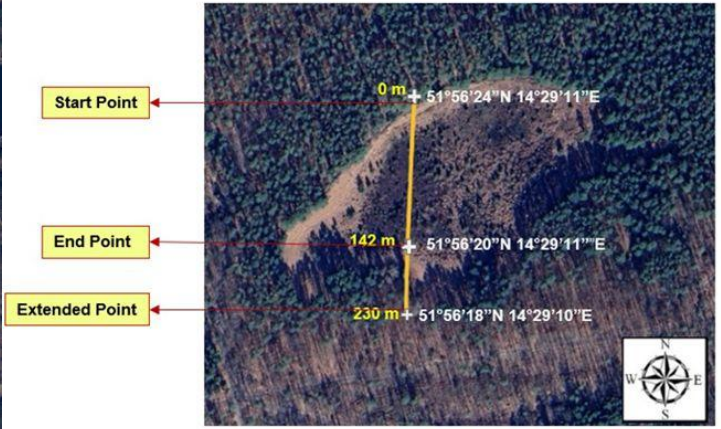
- Civil engineering (e.g. tunnels, foundations)
- Environmental assessments
- UXO detection
- Archaeological investigations

Sounds familiar? …

Weißes Lauch Field Training → Acquire Practical Experience







Setting: Semi-forested area with birch and oak trees, wetlands, and **traces of swamps**

Geological history. Influenced by glacial retreat, hydrological changes, **peat formation**, and climate
effects

The aim was to gain field experience using geophysical methods to study subsurface structures and identify raw materials**.**

The site contains peat layers, sand, and groundwater. all of them are considered natural resources depending on the context (energy, agriculture, ecology).

Weißes Lauch Field Training → Using ERT

This method allowed you to identify variations in subsurface resistivity linked to:

soil composition, moisture, and geological structures





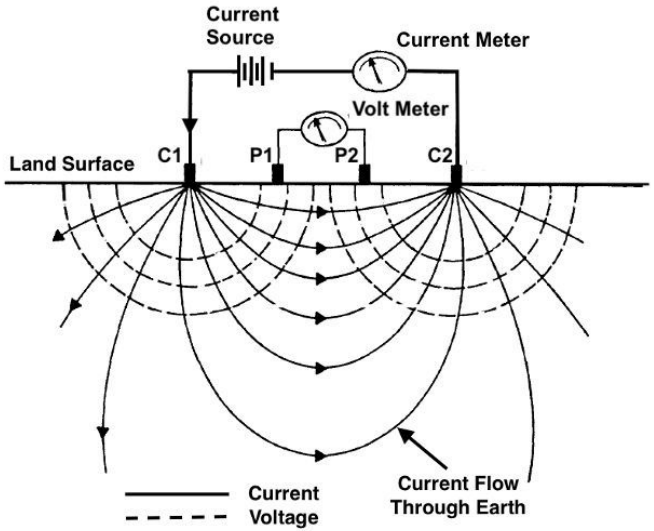








Weißes Lauch Field Training → ERT Studies → How do we estimate the resistivity?



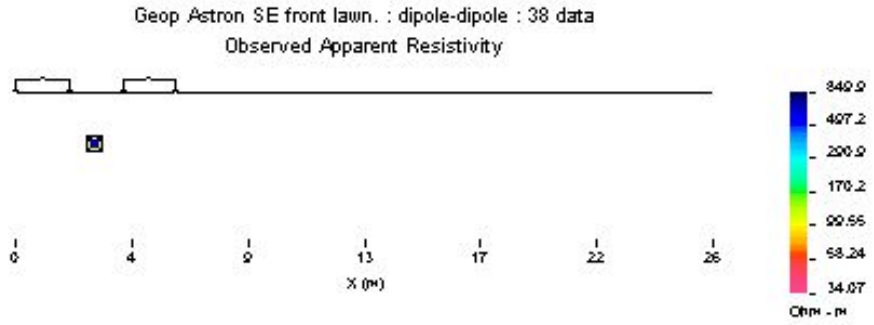
- Determination of the underground situation by measuring of resistivities and comparison with known resistivities of geological materials
- Method is based on *Ohm's Law*

$$R = \frac{V}{I}$$

R: Resistance (Ω)
U: Potential (V)
I: Current (A)

- Uses 4 electrodes, C1 and C2 (current electrodes) and P1 and P2 (potential electrodes)
- Under consideration of the geometric position of the electrodes (K-factor) we obtain the resistivity

Weißes Lauch Field Training → ERT Studies → Apparent Resistivity

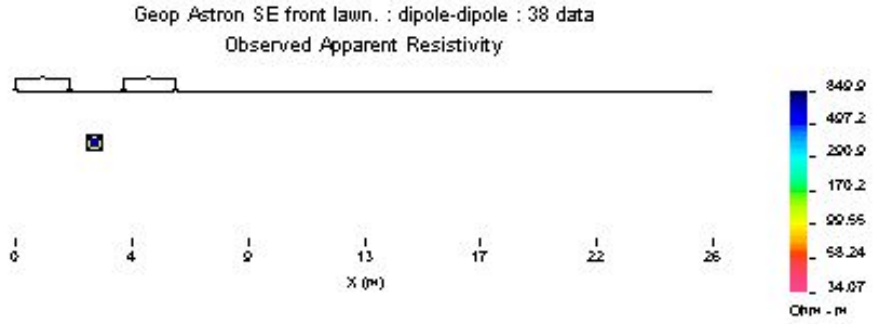


$$\rho_a = k\frac{V}{I}$$

ρ_a: apparent resistivity (Ωm)
k: Geometric factor, represents the distance between the different electrodes
U: Potential (V)
I: Current (A)

It is based on the assumption that the ground is homogeneous and isotropic (homogeneous half space)

Weißes Lauch Field Training → ERT Studies → Apparent Resistivity



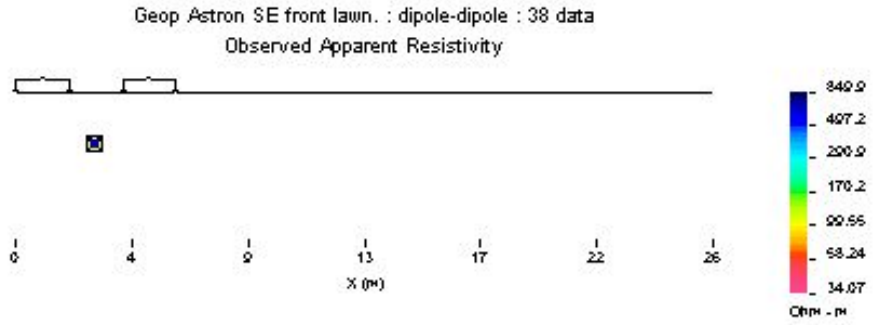
Does this values accurately represents the “true” distribution of subsurface resistivity?

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Weißes Lauch Field Training → ERT Studies → Apparent Resistivity



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It is based on the assumption that the ground is homogeneous and isotropic (homogeneous half space)

Does this values accurately represents the “true” distribution of subsurface resistivity?

That apparent resistivity value would correspond to the true resistivity if the subsurface were homogeneous.

It is a fictitious value, it does not represent the actual resistivity of any specific layer

However, if the ground is layered or has heterogeneities, that value is merely an integrated or weighted average response of the actual resistivities present at depth

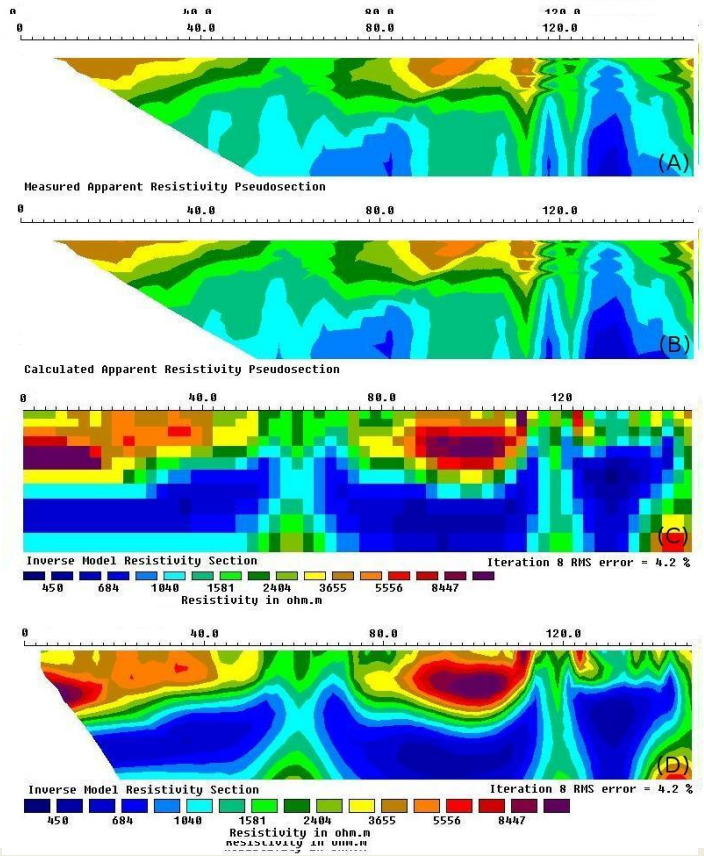
Weißes Lauch Field Training → ERT Studies → Inversion procedure

Measured/Observed resistivity values

Calculated resistivity values

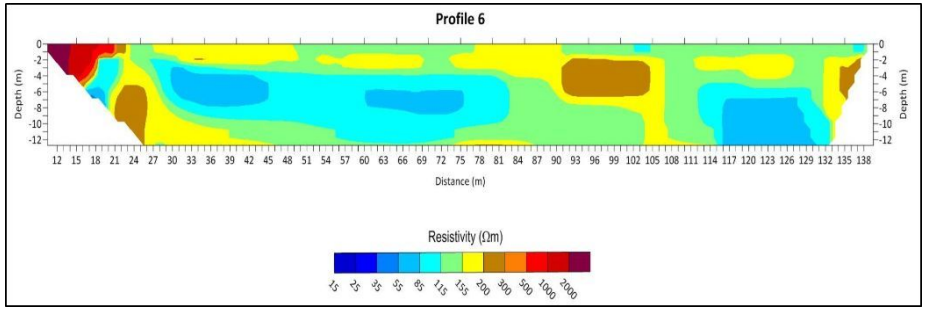
True resistivity values (model blocks)

True resistivity values (contoured section)

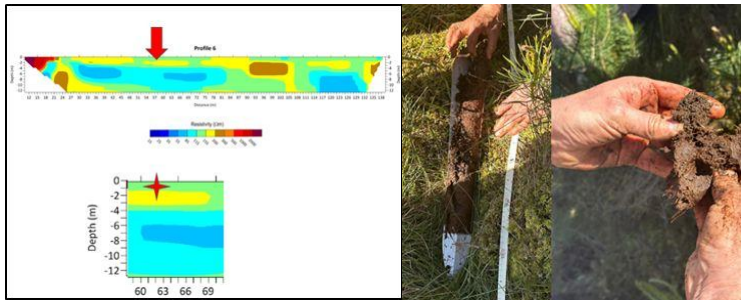


Weißes Lauch Field Training → ERT Studies → Final Resistivity Model

Profiles provided for Interpretation



Drillings at specific locations



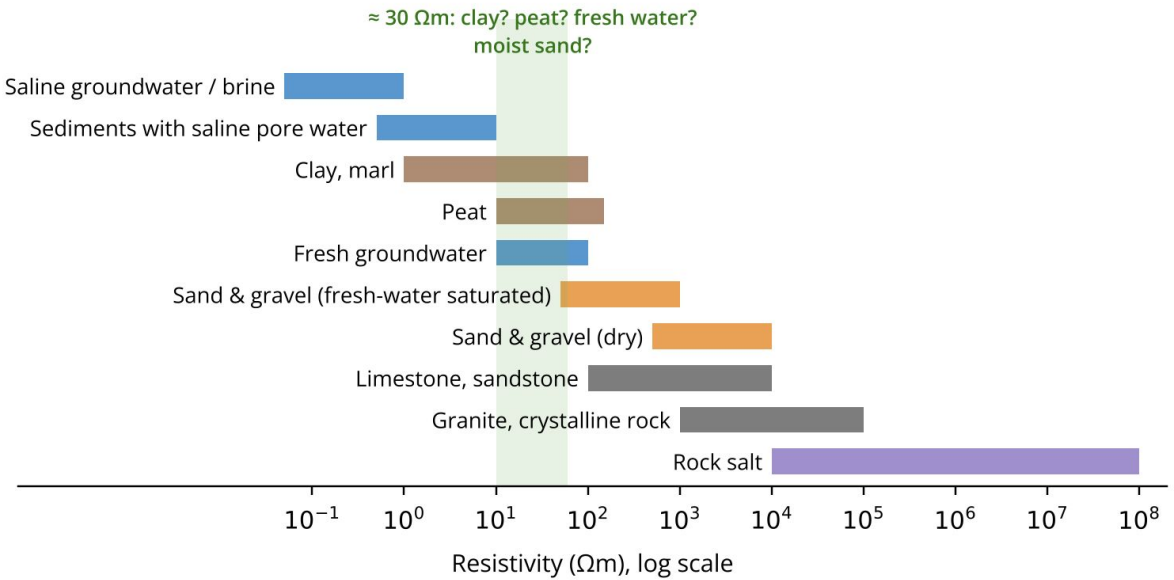
With ERT we measure resistivity, but we don't directly see the material underground. **By drilling at specific points, we can observe what's really there.** whether it's sand, peat, clay, or groundwater.

Once we know that a certain resistivity value corresponds to a particular material at one location, **we can extrapolate** that information to other areas in the profile with similar values. assuming the geological context is consistent.

Material	Resistivity in Ωm
Clay (wet)	3 - 30
Peat (wet)	10 - 85
Sand (wet)	100 - 200
Gravel (wet)	200 - 500
Sand (dry)	> 500

Typical resistivities of local geological materials

Weißes Lauch Field Training → One value, many materials

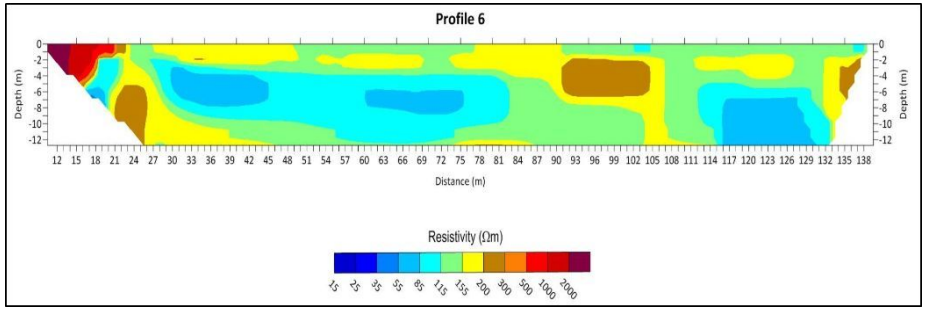


- Resistivity depends on **porosity, saturation, salinity, clay content** and temperature – not only on the material.
- Ranges **overlap**: a 30 Ωm zone could be clay, peat or a fresh-water-bearing sediment.
- That is why you needed the **drillings** at Weißes Lauch to calibrate the model.
- And why, in this course, we **combine methods** that are sensitive to **different properties**.

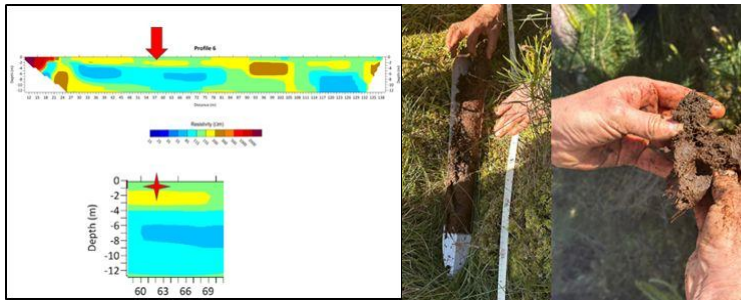
Indicative ranges compiled from Telford et al. (1990) and Reynolds (2011)

Weißes Lauch Field Training → ERT Studies → Final Resistivity Model

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Drillings at specific locations



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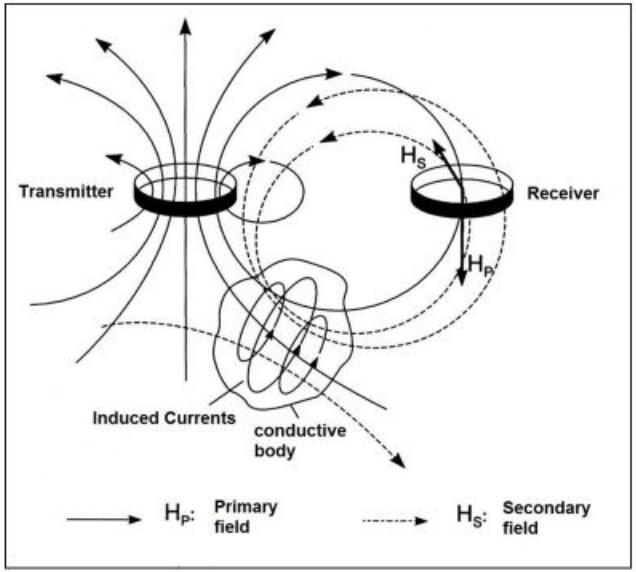
Typical resistivities of local geological materials

So far your experience using this method:

- acquisition
- processing
- interpretation

Weißes Lauch Field Training → EM induction

Electromagnetic (HLEM & VLEM): To measure ground conductivity and infer resistivity indirectly using EM induction







Application of an artificial electromagnetic field, comparison of primary and secondary field, determination of ground conductivity (reciprocal value of resistivity)

$$\frac{H_s}{H_p} \propto \sigma$$



$$\rho = \frac{1}{\sigma}$$

Weißes Lauch Field Training → What it taught us

1

We measure properties, not materials

ERT gives resistivity, not “sand” or “peat”. Ground truth (drillings, samples, outcrops) turns numbers into geology.

2

Apparent ≠ true

Measured apparent resistivities must be inverted. The model is one of many possible solutions (non-uniqueness).

3

One property is rarely enough

Combining methods that see ρ, σ, ε, magnetic susceptibility or γ-activity reduces ambiguity.

4

Survey design decides the result

Electrode spacing, profile length and position control depth and resolution

→ These four ideas are the thread through every method and every case study in this course.

Course Overview

Objective:

- To apply geophysical prospecting methods for the exploration of natural resources and other related applications.

Covering the complete workflow from data acquisition to interpretation.

Contents:

- *Lecture:* Theory about working principles and processing techniques of geophysical prospecting methods.
- *Field training:* Hands-on data acquisition in the field.
- Lab session: Data processing and interpretation.

Focus of the course ⟶ Strong practical orientation

Full workflow:

- From data acquisition to interpretation (diff Weißes Lauch) ⟶ comprehensive approach

Hands-on training

- Survey planning ⟶ selecting appropriate methods based on target, terrain, expected contrasts, and interferences.
- Instrument operation in the field ⟶ use of equipment, quality control.

Data processing

- Commercial software ⟶ powerful, user friendly
- Open-source tools (Python, QGIS) ⟶ flexible toolbox for data management

From Natural Resource Investigation → this course

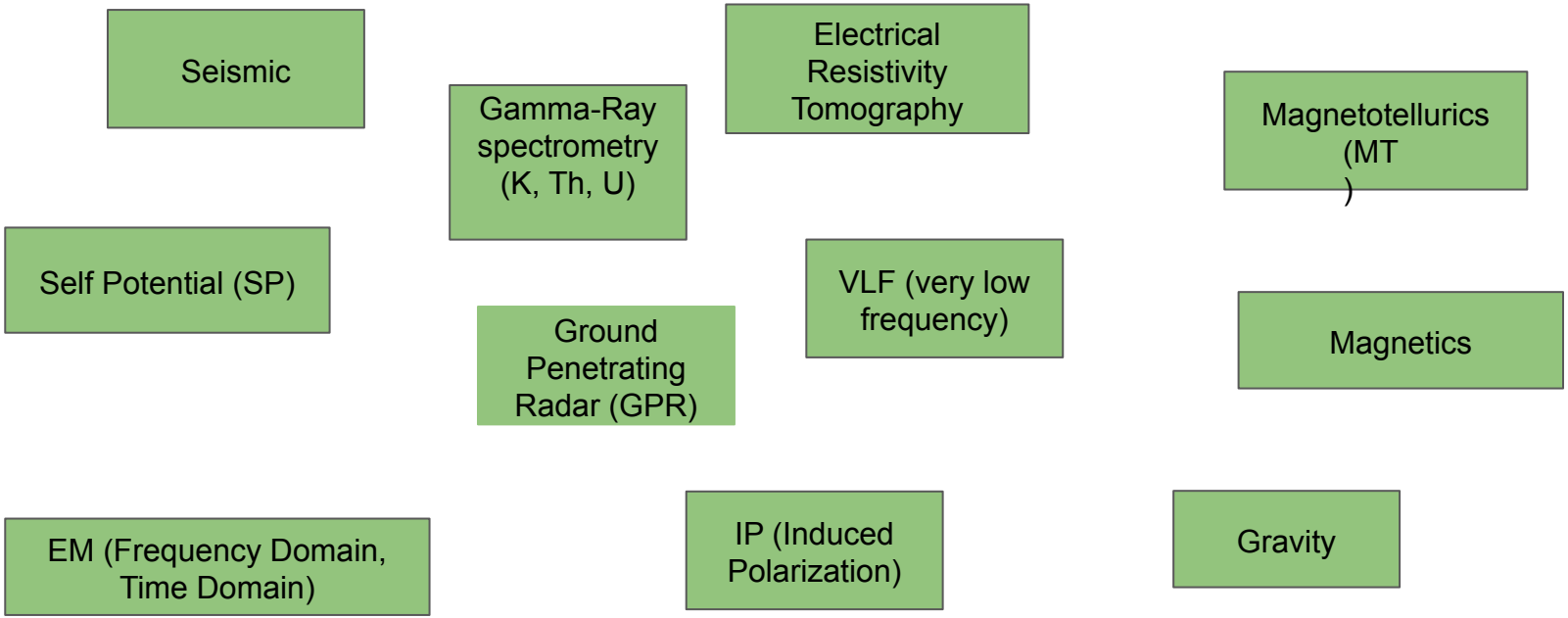
	Natural Resource Investigation (Weißes Lauch)	Advanced Geophysical Methods (this course)
Data	Final resistivity profiles provided for interpretation	You plan the survey and acquire your own data
Methods	ERT, EM induction	+ GPR, magnetometry, radiometry, VLF-EM (ground & airborne)
Workflow	Acquisition and Interpretation	Planning → acquisition → processing → inversion / modelling → interpretation
Scale	One site, a few profiles	From centimetres (GPR) to regional airborne surveys (km^2)
Applications	Peat, sand, groundwater	Groundwater, raw materials, environment, geohazards, archaeology, climate
Output	Interpretation of given models	Technical report + oral presentation of your own field & lab work

Geophysical Methods

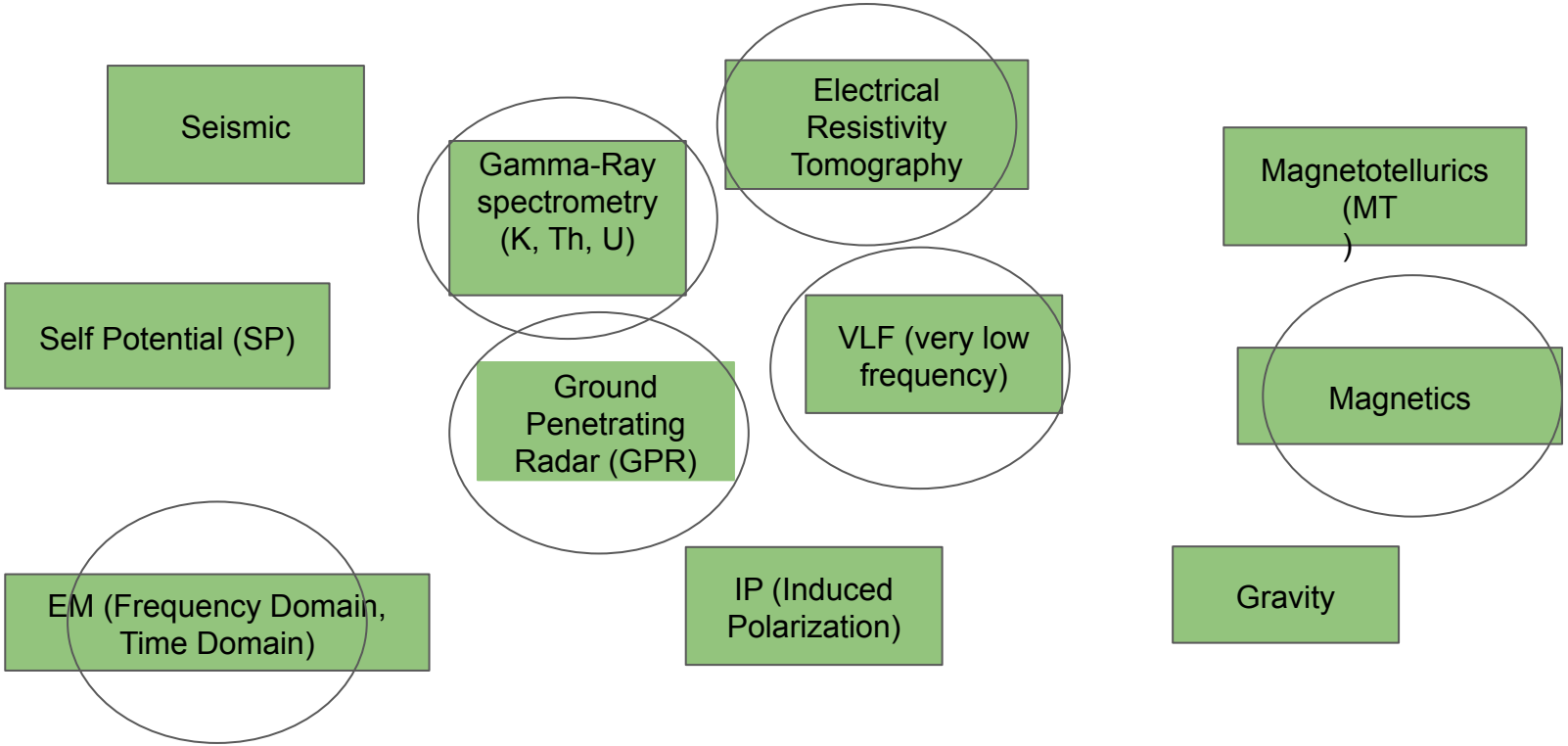
Electrical Resistivity Tomography

EM (Frequency Domain, Time Domain)

Geophysical Methods



Geophysical Methods

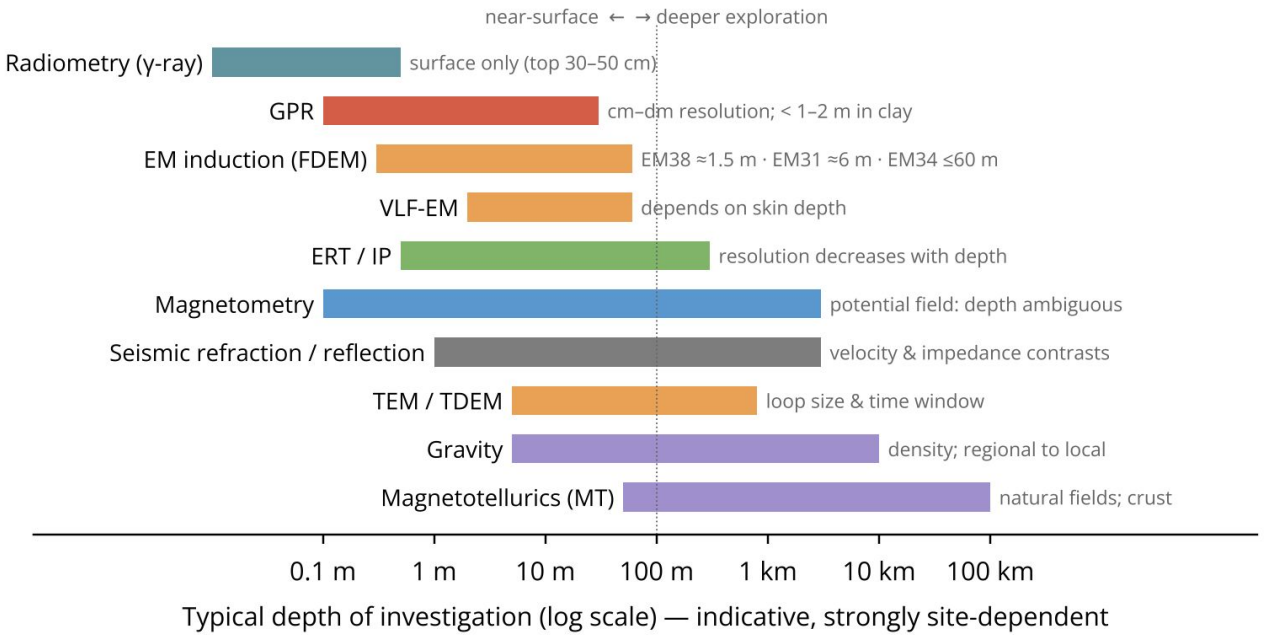


Geophysical Methods → What does each one actually measure?

Method	Physical property	Source	Typical targets
ERT	Electrical resistivity ρ	active – injected current	groundwater, clay, salinity, structures, cavities
IP	Chargeability	active – injected current	clays, disseminated sulphides, some contaminants
EM induction (FDEM / TEM)	Electrical conductivity σ = 1/ρ	active – transmitter coil	salinity, landfills, plumes, conductive ores, metal
VLF-EM	Conductivity contrasts	passive – VLF radio transmitters	water-bearing fractures, faults, shear zones
GPR	Dielectric permittivity ε	active – radar pulse	layers, pipes, foundations, graves, ice, water table
Magnetometry	Magnetic susceptibility, remanence	passive – Earth's field	magnetite-rich rocks, dykes, faults, UXO, kilns
Radiometry (γ-ray)	K, U, Th concentration	passive – natural decay	surface lithology, alteration, U / REE
Seismic	Elastic velocity, impedance	active – hammer, explosives	layering, bedrock depth, rippability
Gravity	Density	passive – Earth's gravity	basins, cavities, dense ore bodies

Rule of thumb: no contrast in the measured property → no anomaly, no matter how good the instrument is.

Geophysical Methods → How deep and how detailed?



- **Depth vs resolution trade-off**: GPR resolves centimetres but only in the first metres; MT reaches kilometres with coarse resolution.
- **Radiometry sees only the surface** (upper 30–50 cm) – it maps soils and outcrops, not depth.
- **Potential fields** (magnetics, gravity) have no fixed depth: amplitude decays with distance and the depth estimate is ambiguous.
- Choose by **target depth AND target size**.

Classification based on deployment method

Geophysical surveys can be conducted:

- from the air
- on the ground
- in marine environments.

The choice depends on factors such a:

- scale
- resolution
- accessibility
- and target depth.

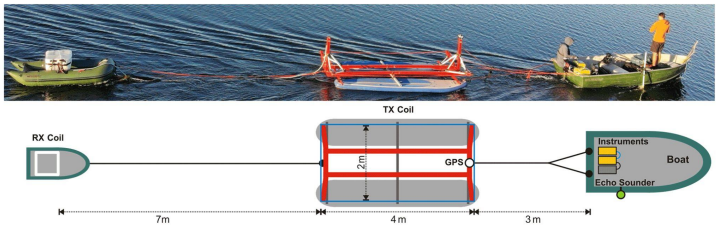












Classification based on deployment method

Airborne methods

- Cover large areas efficiently and rapidly.
- Ideal for regional mapping or inaccessible terrain.
- Common techniques: magnetometry, radiometry, EM, gravimetry.

Ground-based methods

- Offer higher resolution and greater data density.
- Used for detailed site characterization or follow-up investigations.
- Common techniques: ERT, GPR, seismic, magnetics, EM.

Marine-based methods

- Applied in offshore or coastal environments.
- Used for seafloor mapping, resource exploration, and environmental studies.
- Common techniques: marine seismic, magnetics, EM, gravity.

Airborne surveys are used in the early exploration stages,

Ground or marine surveys for detailed investigations.

Methods you will learn in this course

Geophysical methods covered

Ground-based

- **ERT** (subsurface resistivity) → groundwater, structural studies → **Focusing on processing-modelling**
- **EM induction** (electrical conductivity) → environmental and mineral applications → **Focusing on processing-modelling**
- **GPR** (dielectric permittivity and signal reflections) → shallow features like pipes or archaeological remains → **Full workflow : acquisition - processing - interpretation**
- **Magnetometry** (measures magnetic field intensity) → detection of ferrous objects, Unexploded Ordnance (UXO) Detection. → **Full workflow : acquisition - processing - interpretation**

Methods you will learn in this course

Geophysical methods covered

Airborne- based

- **Magnetometry** (measures magnetic field intensity) → regional geological mapping, detection of faults, dikes, mineral exploration (e.g. Fe deposits associated with magnetic minerals suchs a magnetite). → **Acquisition - Processing - Interpretation.**
- **Radiometry** (measures gamma radiation from K, U, Th) → mapping of surface lithology, mineral exploration (e.g, uranium, rare earths, potassic alteration) → **Acquisition - Processing - Interpretation.**
- **VLF-EM** (passive EM response to natural VLF transmitters) → detection of shallow conductors, fracture zones, hydrogeological studies. → **Processing - Interpretation.**

Upcoming classes… 23.10 & 30.10

Goals for the next classes: we start with our first method:

GPR (ground penetrating radar)

Theory

- How GPR works (electromagnetic pulses, reflections)
- Physical principles and signal behavior in different materials
- Applications: shallow structures, archaeology, utilities
- Data processing steps

Acquisition

- Field setup and data collection procedures
- Equipment handling and survey design basics

Lab Session

- Practice with GPR software
- Basic interpretation of profiles

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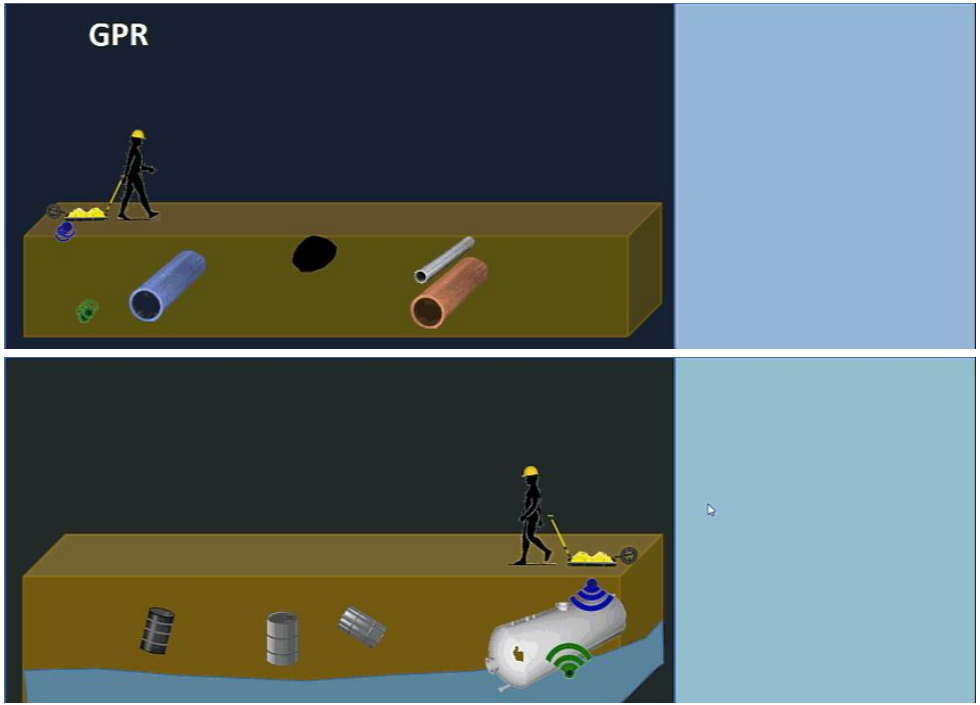
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Acquisition

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Lab Session

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Today class

Real-world problems: Practical applications of geophysical methods

Case studies from the Chair

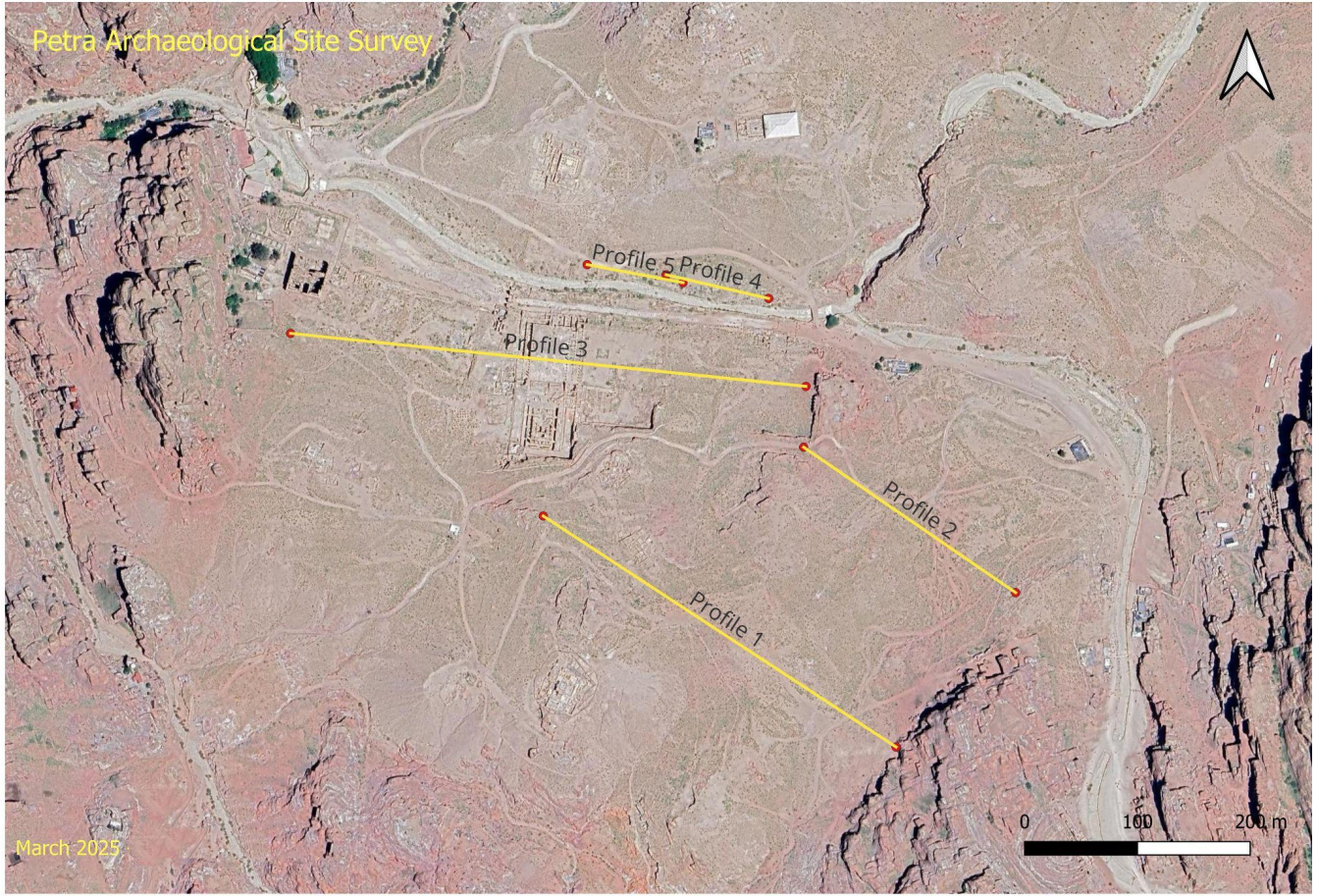
Geophysical ERT Survey in Petra – March 2025



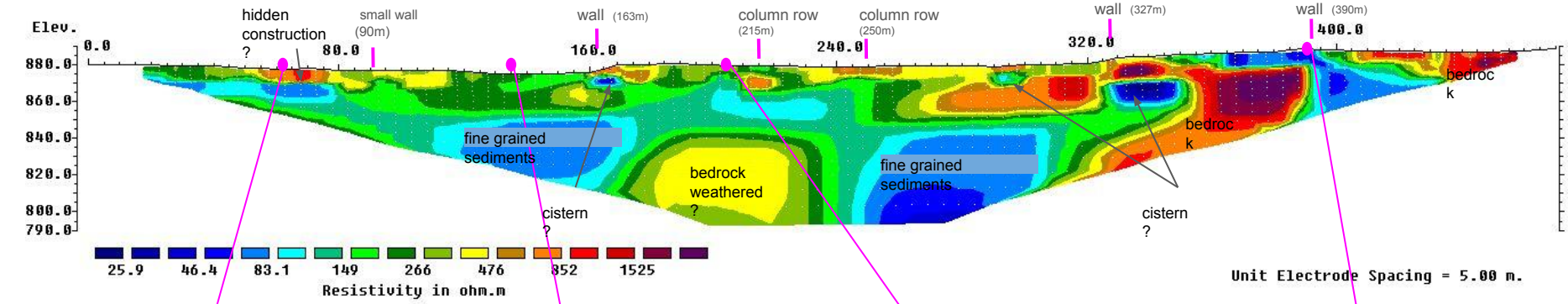




Current Projects of the Chair - Petra Project



Profile 3





forward view at 60 m



forward view at 135 m

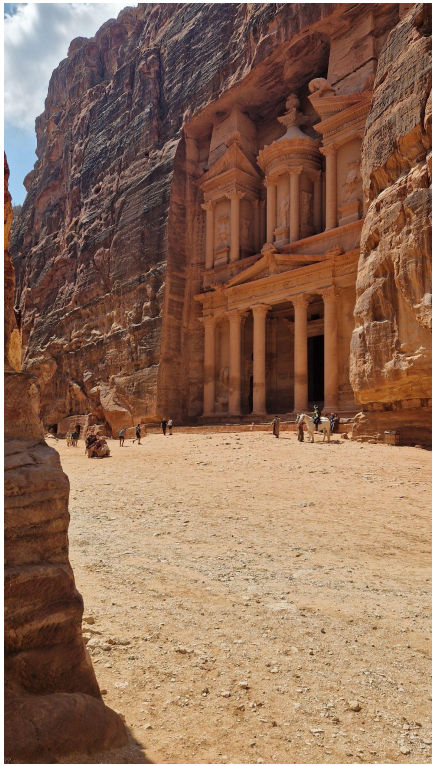


forward view at 203 m



backward view at 390 m

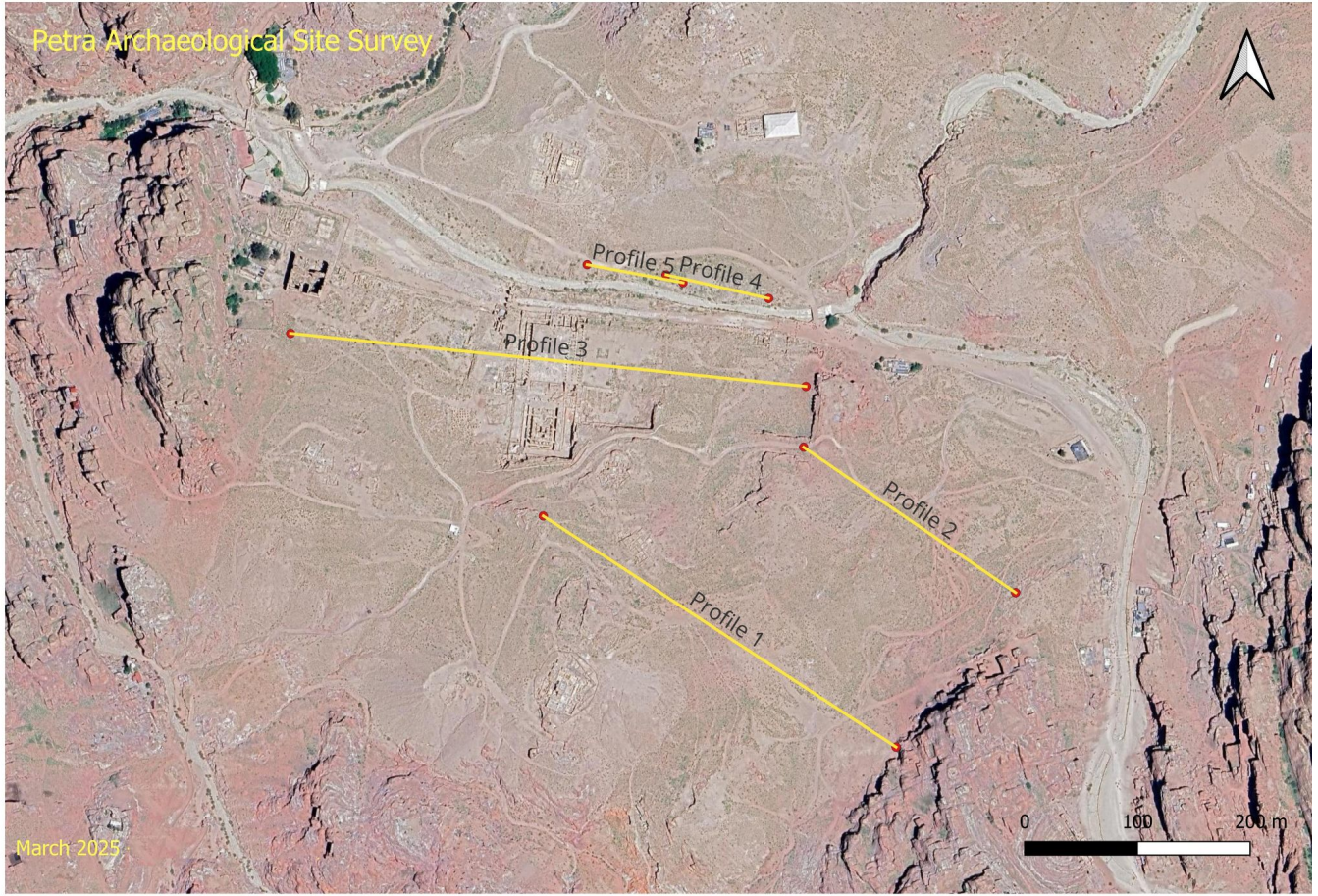
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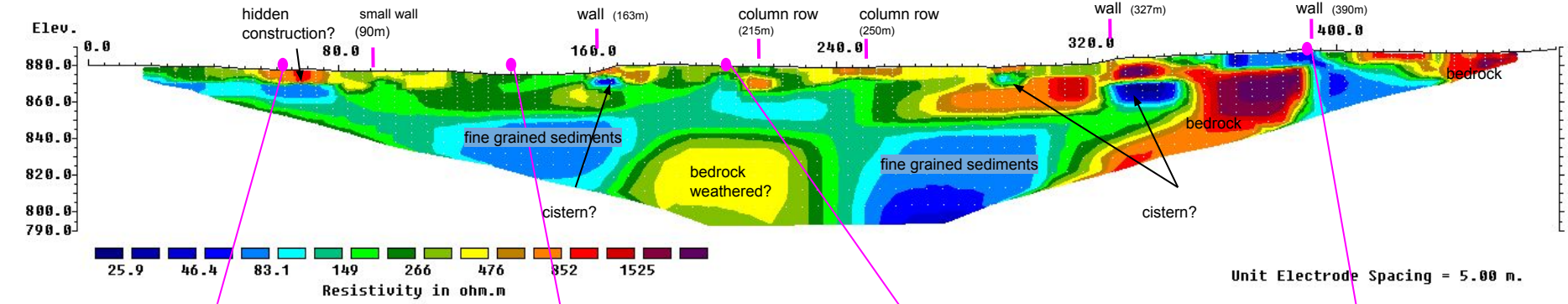




Current Projects of the Chair - Petra Project



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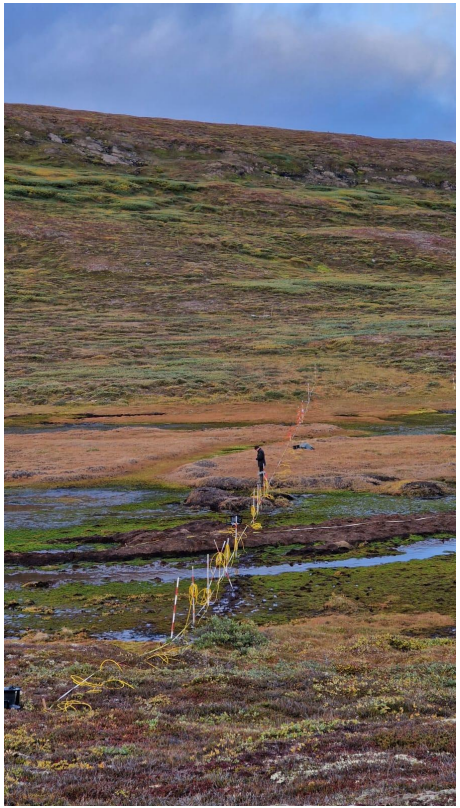


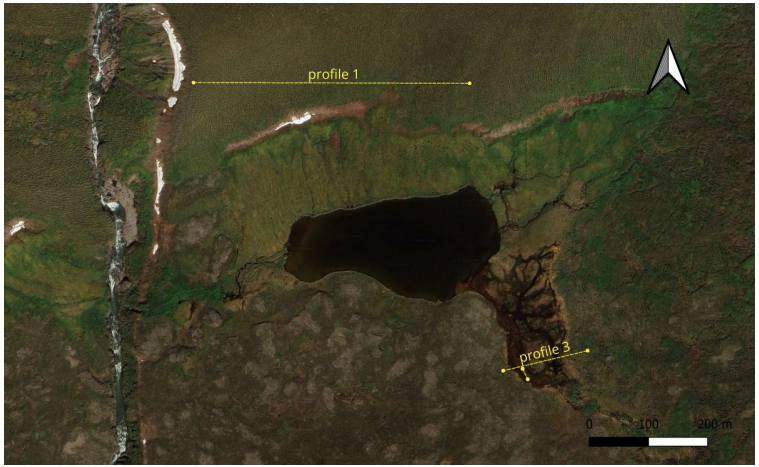
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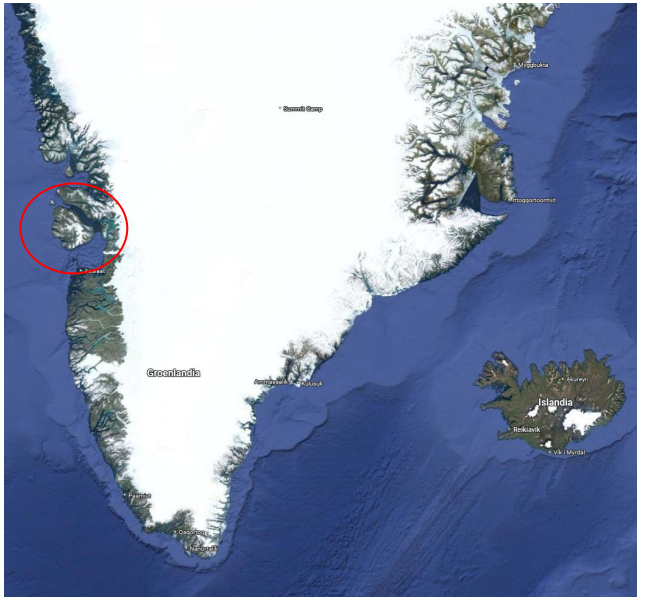


backward view at 390 m

Geophysical ERT Survey Qeqertarsuaq - August 2025



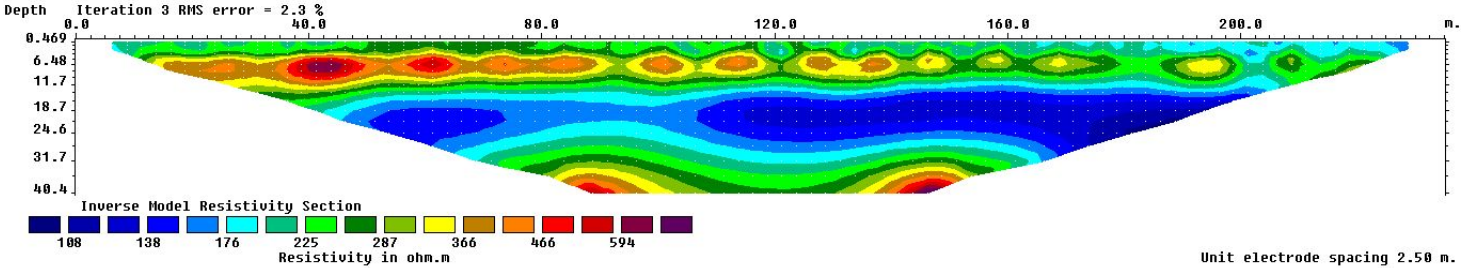




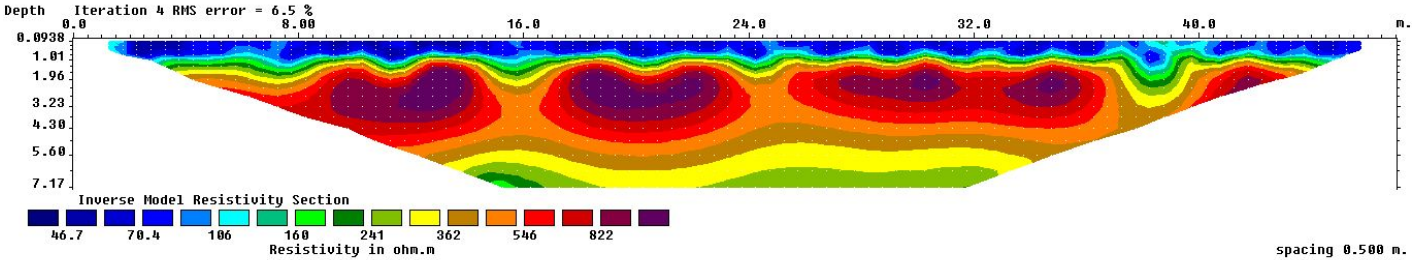




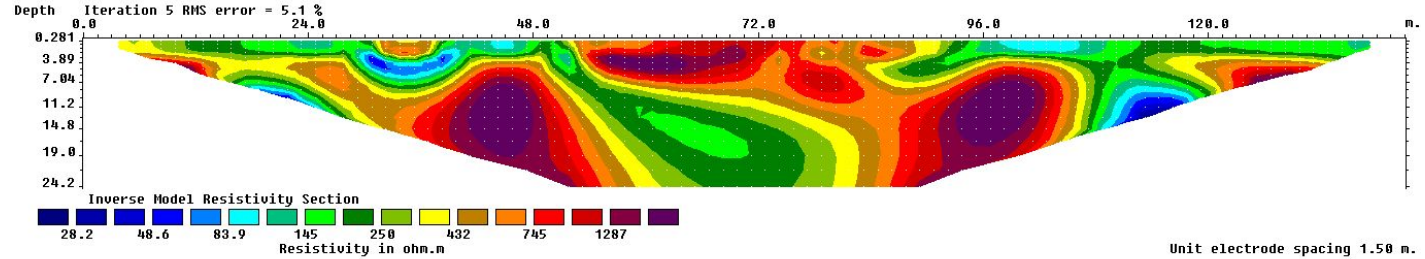
Preliminary results



Profile 1. 5m electrode distance



Profile 1. 1m electrode distance



Profile 3. 3m electrode distance